1. Problem Description

Credit card fraud is a major issue in the financial sector, leading to significant monetary losses for banks and customers. With the rise of online transactions, detecting fraudulent activities has become increasingly complex. Fraudulent transactions are rare compared to legitimate ones, making the dataset highly imbalanced and difficult to model using traditional techniques.

This project aims to develop a machine learning-based system to classify transactions as legitimate or fraudulent using transaction-level data. By analyzing patterns such as transaction amount, time, location, and user behavior, the system can help identify suspicious transactions efficiently. The model focuses on maximizing fraud detection while minimizing false negatives, ensuring better security in real-world applications.

Dataset Description

The dataset used in this project is a synthetic credit card transaction dataset containing around 100 transactions. Each row represents a single transaction with features such as transaction amount, transaction time, transaction type, location, device type, and fraud history.

The dataset includes both numerical and categorical features, which were preprocessed through encoding and scaling. The target variable is binary:

- **0 → Legitimate Transaction**
- **1 → Fraudulent Transaction**

To simulate real-world conditions, the dataset was modified to include overlapping patterns and slight noise, ensuring realistic model behavior.

Flow of the System:

Data Collection → Data Preprocessing → Train-Test Split → Scaling → SMOTE (Imbalance Handling) → Model Training → Model Evaluation → Prediction Output

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Modules for Project Implementation

Modules Used

- **NumPy**: Used for numerical computations and array operations.
- **Pandas**: Used for data loading, cleaning, and manipulation.
- **Matplotlib**: Used for plotting graphs such as ROC curve.

- **Scikit-learn**: Used for machine learning models, preprocessing, and evaluation metrics.
- **Imbalanced-learn (SMOTE)**: Used to handle imbalanced dataset by generating synthetic samples.

Platform Used

The project was implemented using **Jupyter Notebook**, which provides an interactive environment for writing, executing, and visualizing Python code. It allows step-by-step execution and easy debugging, making it suitable for machine learning workflows.

Machine Learning Model / Searching Technique Used

The following supervised machine learning algorithms were used:

- **Logistic Regression**: A linear classification model used for binary classification based on probability.
- **K-Nearest Neighbours (KNN)**: A distance-based algorithm that classifies data points based on nearest neighbors.
- **Support Vector Machine (SVM)**: A model that finds the optimal hyperplane to separate classes effectively.

These models were trained on the processed dataset and compared based on performance metrics.

Evaluation Metrics Used

The models were evaluated using the following metrics:

- **Accuracy**: Measures overall correctness of the model.
- **Precision**: Measures how many predicted fraud cases are actually fraud.
- **Recall**: Measures how many actual fraud cases are detected (most important metric).
- **F1-Score**: Harmonic mean of precision and recall.
- **Confusion Matrix**: Displays true positives, false positives, true negatives, and false negatives.

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5. Results and Visualizations

The models were trained and evaluated on the dataset, and the following observations were made:

- Logistic Regression and SVM showed strong performance with high accuracy.
- KNN showed comparatively lower accuracy due to sensitivity to data distribution.
- The confusion matrix helped visualize correct and incorrect predictions.
- The ROC curve demonstrated the model's ability to distinguish between classes effectively.

The results indicate that the system can successfully identify fraudulent transactions with reasonable accuracy.

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6. Conclusion and Future Scope

This project successfully developed a machine learning-based credit card fraud detection system. The pipeline included data preprocessing, imbalance handling using SMOTE, model training, and evaluation using multiple metrics. The models were able to classify transactions effectively and demonstrate realistic performance.

Future Scope:

- Use larger real-world datasets for better accuracy.
- Implement advanced models like Random Forest or Neural Networks.
- Deploy the model as a real-time fraud detection system.
- Integrate with banking systems for live monitoring.

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7. References

1. Python Documentation – https://docs.python.org
2. Scikit-learn Documentation – https://scikit-learn.org
3. Kaggle Datasets – https://www.kaggle.com
4. Imbalanced-learn Documentation – https://imbalanced-learn.org
5. GeeksforGeeks – https://www.geeksforgeeks.org